

01 variables

干汝宣

1.variables

number: integer, float

boolean (True/False)

string: characters

none: used to define no value

使用type()函数可查看变量类型，len()函数查看字符串长度

- 换行： `"""hello world"""`

不换行：

```
"hello \  
world"
```

- 字符串不可双引号内套双引号，若需使用引号，有如下方法

```
text1 = 'She said, "What time is it?'"  
text1 = "She said, \"What time is it?\""
```

NOTE:

string cannot begin with number

use meaningful names

use lowercase and _

2.functions about string

- 字符串合并（无空格） `string3 = string1 + string2`

- 提取第n位 `text[n]`

0表示左数第一位，-1表示右数第一位，1:6表示1至5位，:6是省略0，1: 是省略最右位

- 变成小写 `" Python BootCamp".lower()`

```
>>> python bootcamp
```

- 去掉空格 `" Python BootCamp".strip()`

```
>>> Python BootCamp
```

- 数有几个 `" Python BootCamp is a Python program!".count("Python")`

```
>>> 2
```

- **第一个出现的位置** " Python BootCamp is a Python program!".find("Python")
 >>> 1
 | 左数第一位是0，返回-1代表没有
- **是否以P起始** "Python BootCamp".startswith("P")
 >>> True

3.arithmetic operators

Operator	Operation
+	Addition
-	Subtraction
*	Multiplication
/	Division
**	Exponent
//	Floor division

Operator	Description	Example
==	equal	1 == 1
!=	not equal	2.3 != 2.313
<	less than	2.5 < 9
>	greater than	2.4 > 2.399
<=	less than or equal	3 <= 3
>=	greater than or equal	2.4 >= 2.399

Operator	Description
x or y	if either of x or y is True, return True; otherwise, return False
x and y	if both x and y are True, return True; otherwise, return False
not x	reverse x and return (return False, if x is True)

4.ways to format string

```
print(name, "has", str(heads), "heads and", str(arms), "arms.")  
print(name + " has " + str(heads) + " heads and " + str(arms) + " arms.")  
print(f"{name} has {heads} heads and {arms} arms.")  
("{} has {} heads and {} arms.".format(name, heads, arms))  
"%s has %s heads and %s arms." % (name, heads, arms))
```